MILITARY SPECIFICATION

VISORS, NEODYMIUM LASER, PROTECTIVE FOR AIRCREWMAN'S HELMET

This specification is approved for use by the Naval Air Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements for hard coated polycarbonate laser protection visor lenses for the EEK-3/P dual visor assembly.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. Unless otherwise specified, the following specifications, standards, and handbooks of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

L-P-393	-	Plastic Molding Material, Polycarbonate, Injection and Extrusion.
PPP-B-636	-	Boxes, Shipping, Fiberboard.
PPP-B-676	-	Boxes, Setup.

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MIL-E-12397	-	Eraser, Rubber-pumice (For Testing Coated Optical Elements).
MIL-H-85047	-	Helmet Assembly, HGU-34/P.

STANDARDS

FEDERAL

FED-STD-406	-	Plastic, Methods of Testing.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Engineering Specifications and Standards Department (Code 93), Naval Air Engineering Center, Lakehurst, NJ 08733, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	-	Marking for Shipment and Storage.
MIL-STD-130	-	Identification Marking of U.S. Military Property.
MIL-STD-810	-	Environmental Test Methods

2.1.2 Other Government documents, drawings, and publications. The following other Government documents form a part of this specification to the extent specified herein.

DRAWINGS

US ARMY ARMAMENT RESEARCH AND DEVELOPMENT COMMAND

7680606	-	Tester, Abrasion, Optical Coating.

NAVAL AIR SYSTEMS COMMAND

765AS315	-	Lens Subassembly, Neodymium, Inner, Dual Visor, EEK-3/P.
765AS316	-	Lens, Inner, Neodymium Laser, Protective, Dual Visor, EEK-3/P.
765AS292	-	Holder, Neodymium Laser Protective Lens, Aircrewman's Helmet.
765AS293	-	Guide, Neodymium Laser Protective Lens, Aircrewman's Helmet.
765AS294	-	Pin, Pivot, Neodymium Laser Protective Lens Holder, Aircrewman's Helmet.

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. The issues of the documents which are indicated as DoD adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ANSI/ASTM D 412	-	Rubber Properties in Tension.
ANSI/ASTM D 790	-	Plastic and Electrical Insulating Materials, Properties of.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

(Industry association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

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3. REQUIREMENTS

3.1 Qualification. The visor furnished under this specification shall be products which are qualified for listing on the applicable qualified products list at the time set for opening of bids (see 4.3 and 6.3).

3.2 First article. When specified, a sample shall be subjected to first article inspection (see 4.4 and 6.4).

3.3 Materials. Materials shall conform to applicable specifications and shall be as specified herein and on applicable drawings. Materials which are not covered by specifications, or which are not specifically described herein, shall be of the best quality, of the lightest practicable weight, and suitable for the purpose intended.

3.3.1 Visors. The plastic material used for molding the visors shall be in accordance with L-P-393.

3.4 Design and construction. The design and construction of the visor shall be in accordance with Drawings 765AS315, 765AS316, 765AS292, 765AS293 and 765AS294 (see 6.2.1c).

3.5 Performance.

3.5.1 Light transmittance. The light transmittance of the visor, when calculated as specified in 4.7.2, shall be not less than 55 percent throughout the critical area. The transmittance of point "C" for the left eye and point "C" for the right eye of the same visor shall not differ by more than 20 percent. The total visible transmittance at the top of the visor shall not exceed the total transmittance at the bottom of the visor.

3.5.2 Neutrality. The spectral transmittance curve of the visor, when calculated as specified in 4.7.3, may vary between 430 and 730 nanometers (nm). The average percentage deviation within nine spectral bands shall be less than 30 percent.

3.5.3 Chromaticity. The chromaticity coordinates X and Y of the visor, when computed as specified in 4.7.4, shall be within the limits indicated in figure 1.

3.5.4 Diffuse transmittance. The diffuse transmittance of the visor, when determined as specified in 4.7.5, shall not exceed 5 percent of the total visible transmittance.

3.5.5 Ultraviolet transmittance. The ultraviolet transmittance of the visor, when determined as specified in 4.7.6, shall be less than 5 percent in the erythemal band at 290 to 320 nanometers.

3.5.6 Infrared transmittance. The infrared transmittance of the visor, when measured as specified in 4.7.7, shall be 0.001 or less (optical density 3.0 or greater) across the entire surface of the visor at 1064 nanometers wave length.

3.5.7 Prismatic deviation.

3.5.7.1 Vertical prismatic deviation. The vertical prismatic deviation between point "C" for the right eye and point "C" for the left eye, when measured as speci-

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fied in 4.7.8, shall not differ by more than 0.25 diopters. In addition, there shall be no abrupt transitions in prismatic deviation throughout the critical and semi-critical area of the visor.

3.5.7.2 Horizontal prismatic deviation. The sum of the horizontal prismatic deviation at point "C" for the right eye and point "C" for the left eye, when measured as specified in 4.7.8, shall not exceed 0.75 diopters. The difference between the horizontal prismatic deviation at point "C" for the right eye and point "C" for the left eye shall not exceed 0.18 diopters. In addition, there shall be no abrupt transitions in prismatic deviation throughout the critical and semi-critical area of the visor.

3.5.8 Spherical and cylindrical power. The spherical and cylindrical power, when measured as specified in 4.7.9, shall not exceed plus or minus 0.125 diopters at point "C".

3.5.9 Critical vision area distortion. The distortion within the critical vision area, when examined as specified in 4.7.10, shall be within the acceptable limits shown in figure 2.

3.5.10 Windblast resistance. The visors, when tested as specified in 4.7.11, shall remain in the full down and locked position and there shall be no visible evidence of material failure.

3.5.11 Haze. The haze value of the visor shall not exceed 2.0 percent when tested as specified in 4.7.12.

3.5.12 Abrasion resistance. After being subjected to the abrasion tests specified in 4.7.13, the increase in haze shall not exceed 6 percent and the decrease in transmittance shall not exceed 4 percent.

3.5.13 Adhesion of coating. The coating, when tested as specified in 4.7.14, shall not be removed, dislodged, or affected in any way. Removal or loosening of the coating shall be cause for rejection.

3.5.14 Crack propagation resistance (flexural strength). When tested as specified in 4.7.15, the visor specimens shall exhibit a minimum flexural strength of 11,000 psi. This level shall be achieved without cracking or reaching the maximum fiber strain of 5 percent.

3.5.15 Impact resistance. The visor, when tested as specified in 4.7.16, shall not fracture. A visor shall be considered to have fractured if it cracks (through its entire thickness) into two or more separate pieces or if any material visible to the naked eye is detached from the ocular surface.

3.5.16 Penetration resistance. The visor shall not be pierced when subjected to the penetration test specified in 4.7.17.

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3.6 Identification of product. Each visor shall be identified in accordance with MIL-STD-130 and the applicable drawing.

3.7 Workmanship. All visors shall be constructed to produce an item free from all defects which would affect proper functioning in service. The visors shall conform to the quality and grade of product established by this specification. The occurrence of defects shall not exceed the acceptance criteria established herein.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspection. The inspections specified herein are classified as follows:

a. Qualification inspection. Qualification inspection consists of tests and examinations performed on samples submitted for approval as qualified products (see 4.3).

b. First article inspection. First article inspection consists of tests and examinations performed on samples which are representative of the production item after award of contract to determine that the production item conforms to the requirements of this specification (see 4.4).

c. Quality conformance inspection. Quality conformance inspection consists of tests and examinations performed on individual products or lots with the requirements set forth in this specification (see 4.5).

4.3 Qualification inspection. Qualification inspection shall consist of the examinations and tests specified in table I.

4.3.1 Qualification inspection samples. The qualification inspection samples shall consist of four visors. The qualification samples shall be forwarded as specified in 6.3. The samples shall be plainly identified by securely attached tags marked with the following information:

> Sample submitted by (name) (date) for qualification inspection in accordance with the requirements of MIL-V-22272D(AS) under authorization (reference authorizing letter and number) (see 6.3).

4.3.2 Retention of qualification. The retention of qualification shall consist of verification to determine compliance of the qualified visor with the requirements

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of this specification. Every two years each manufacturer shall be requested to forward to NADC certification signed by a responsible official of management, attesting that the listed product(s) is still available from the listed plant, and can be produced under the same conditions as originally qualified; i.e., same process, materials, construction, design, manufacturer's part number, or designation; and meets the requirements of the current issue of the specification. Failure to provide certification will be cause for removal from the QPL. After completion of the certification review, the QPL shall be reprinted to show the date of validation.

4.4 First article inspection. First article inspection shall consist of the following examinations and tests:

Visual and dimensional examination

Packaging and identification

Light transmittance

Neutrality

Diffuse transmittance

Prismatic deviation

Critical vision area distortion

4.4.1 First article samples. Unless otherwise specified, as soon as practicable, after the award of the contract or order, the manufacturer shall submit four visors. The samples shall be representative of the construction, workmanship, components, and materials to be used during production. When a manufacturer is in continuous production of these units from contract to contract, submission of further first article samples on the new contract may be waived at the discretion of the acquiring activity (see 6.2.1d). Approval of the first article samples or the waiving of the first article inspection does not exclude the requirements of submitting to the quality conformance inspection. The first article inspection samples shall be furnished to the Government as directed by the contracting officer (see 6.2.1e).

4.4.1.1 First article information. Upon completion of the first article inspection, all the applicable inspection reports, recommendations and comments pertinent for use in monitoring production shall be recorded (DID No. DI-T-5329) and will be forwarded by the Government activity responsible for the inspection program (see 6.2.1e) to the contracting officer. One visor shall be returned to the manufacturer for use in monitoring production. The other visors will be consumed or destroyed in the first article inspection and shall not be considered as part of the quantity to be delivered under contract.

4.5 Quality conformance inspection. The sampling and inspection levels shall conform to MIL-STD-105. Quality conformance inspection shall consist of the following:

Visual and dimensional examination

Packaging and identification

Light transmittance

Infrared transmittance

Critical vision area distortion

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4.5.1 Sampling.

4.5.1.1 Inspection lot.

4.5.1.1.1 Visors. An inspection lot shall be expressed in units of visors made under essentially the same conditions and from the same materials. The sample unit shall be one visor.

4.5.1.1.2 Packaging. An inspection lot shall be expressed in units of one fully prepared shipping container, containing visors, fully prepared for delivery, made from essentially the same materials and components. The sample unit shall be one shipping container, containing visors, fully prepared for delivery with the exception that it need not be sealed.

4.5.1.2 Sampling for tests and examinations of the visors. The sample size, acceptance criteria, tests and examinations required for the visors shall be as specified in table II.

4.6 Test conditions.

4.6.1 Cleanliness. Before and after subjecting the visor to any test, the visor shall be thoroughly and carefully cleaned to remove dirt, oil, fingermarks, etc., using a mild detergent and water and dried with a soft cloth or lens tissue in order to assure that the following tests are performed on a clean visor.

4.7 Inspection methods.

4.7.1 Visual and dimensional examinations. Every visor shall be examined visually for critical defects to determine conformance with this specification. Each visor, selected as a sample unit from the lot, shall be thoroughly checked dimensionally and examined visually for minor defects to determine conformance to this specification. Table III shall be used to classify the defects found.

4.7.1.1 Packaging and identification. Each of the fully prepared shipping containers, containing visors, selected as a sample unit from the lot, shall be visually examined to determine that the packaging, packing and marking conform to this specification. The classification of defects, table IV, shall be used to classify the defects found.

4.7.2 Light transmittance. A spectrophotometric curve shall be obtained for a visor lens sample. The daylight transmittance, Tc, shall be calculated as shown in table VI. The transmittance shall be measured with a CIE source C which approximates average daylight. An example of a type C source is a gas-filled tungsten lamp calibrated at 2854°K and an appropriate set of filters. The combination of lamp and filters shall have a combined color temperature of approximately 6500°K. A detector, with cell response matched to the standard luminosity curve of the eye, shall be used to measure illuminance with and without the visor lens. The transmittance shall be computed as the percent ratio of the illuminance measured with the visor lens to the illumance measured without the visor lens. The visor shall pass the requirements specified in 3.5.1.

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4.7.3 Neutrality. The neutrality shall be calculated from the spectrophotometric values using the Judd Daylight Duplication method (shortened form). See example, table V. The visor shall pass the requirements specified in 3.5.2.

4.7.4 Chromaticity. The chromaticity shall be calculated from the spectrophotometric values. See example, table VI. The visor shall pass the requirements specified in 3.5.3.

4.7.5 Diffuse transmittance. The diffuse transmittance shall be determined in accordance with FED-STD-406, method 3031. Diffuse transmittance may also be measured on the Hardy GE spectrophotometer by taking a curve with the sample placed against the window of the integrating spheres and another curve with the sample placed at the entrance to the transmission compartment; the difference between these two curves defines the percent diffuse transmittance of the sample. The Cary spectrophotometer with the diffuse reflectance accessory may also be used. The visor shall pass the requirements specified in 3.5.4.

4.7.6 Ultraviolet transmittance. The ultraviolet transmittance shall be determined with an appropriate spectrophotometer (Cary, Beckman DK, etc.). Transmittance readings shall be obtained at 290, 300, 310, and 320 nanometers and these readings shall be summed and then divided by four to obtain the average ultraviolet transmittance of the sample. The visor shall pass the requirements specified in 3.5.5.

4.7.7 Infrared transmittance. The infrared transmittance at 1064 nm shall be measured by using a neodymium laser or other suitable source which radiates at 1064 nm and a laser power meter, bolometer or similar detecting device. The optical density or transmittance recorded shall pass the requirements specified in 3.5.6.

4.7.8 Prismatic deviation. The prismatic deviation both vertically and horizontally shall apply to readings taken on the visor "as worn" (1.70 inches from the concave surface), when measured at point "C". The test equipment shall be a calibrated transit or telescope, a lens holder, and an appropriate illuminated target at a 35-foot distance or equivalent equipment. The visor shall meet the requirements specified in 3.5.7.

4.7.9 Spherical and cylindrical power. The spherical and cylindrical power shall be determined with the same apparatus used for measuring prismatic deviation, if the test equipment is appropriate, or shall be determined with a special lensometer or vertometer which measures from -0.25 to +0.25 diopters with 0.01 divisions (instead of the usual lensometer or vertometer which measures from -20.0 to +20.0 diopters with 0.12 divisions). The visor shall meet the requirements specified in 3.5.8.

4.7.10 Critical vision area distortion. With the visor inserted into the apparatus described in figure 3 and its surface normal to the line of sight, examine for distortion. The degree of off-parallelism will constitute the amount of distortion and this distortion pattern shall be compared with figure 2. The visor shall meet the requirements specified in 3.5.9.

4.7.11 Windblast resistance. The laser protective visor lens shall be installed in a PRU-36/P dual visor assembly in place of the inner track clear lens. The visor assembly shall be installed on a large size PRK-37/P helmet shell assembly with PRK-40/P helmet liner. The helmet shell and liner shall conform to MIL-H-85047. The

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helmet, visor and liner assembly, and a MBU-12/P oxygen mask assembly shall be fitted to a head form securely mounted to a test platform. The inner laser protective lens shall be placed in a full down and locked position. The outer neutral lens shall be placed in a full up and locked position. There shall be a minimum gap between the laser protective lens and the oxygen mask.

The helmet assembly shall be subjected to a three second duration windblast, at a peak velocity of 450 knots equivalent airspeed (KEAS), (520 MPH), at each attitude listed below. The rise time from 0 to 450 KEAS shall be 80 ± 20 milliseconds (ms), and the velocity shall decay from 450 KEAS to 260 KEAS in approximately 3 seconds.

1st Blast	Head-On
2nd Blast	Yaw 45° Right
3rd Blast	Pitch 45° Aft
4th Blast	Pitch 45° Aft, and Yaw 45° Left.

With the helmet secured, so that it will not tear away from the head form, the visor shall pass the requirements specified in 3.5.10.

4.7.12 Haze. The haze shall be determined in accordance with FED-STD-406, method 3022. The visor shall pass the requirements specified in 3.5.11.

4.7.13 Abrasion resistance. The haze and luminous transmittance of the coated specimens shall be determined before and after the abrasion test as specified in FED-STD-406, method 3022. The abrasion test shall be performed by rubbing with a specially standardized eraser mounted as illustrated in figure 4 and held approximately normal to the visor. The device and eraser material indicated in figure 4 shall be used for this test. The eraser shall be rubbed across the surface of the coated visor from one point to another, over the same path, for 20 complete cycles with a force of 2.0 to 2.5 pounds continuously applied. Wherever possible, rubs of about 1 inch length are preferred. After the rubbing has been completed, the lens shall be thoroughly cleaned to remove dirt, film, fingermarks and grease marks using a mild detergent and water followed by drying with a soft cloth or lens tissue. The visor shall meet the requirements of 3.5.12.

4.7.14 Adhesion test method. The specimens shall be subjected to humidity tests as specified in MIL-STD-810, method 507, for 120 hours (five cycles). Immediately upon removal from the humidity chamber, the samples shall be dried with a cotton cloth and then subjected to the adhesion test. A test tape, with an adhesion rating of 40 ounces per inch of width, shall be firmly applied to the front surface of the visor. The area covered shall be 2 square inches. The tape is then removed using a snapping motion applied 90° to the specimen. Two trials at separate locations shall be made using this method. The following 3M tapes are acceptable: 670 (preferred), 202, 218 and 232. The test sites shall be closely examined. The visor shall meet the requirements of 3.5.13.

4.7.15 Crack propagation resistance (flexural strength).

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4.7.15.1 Specimen. The test specimen shall be taken from a completely formed visor and from an area to provide as flat a sample as possible. The specimen shall be a rectangular bar, with the same thickness (t) as that of the visor. The width shall be 1/2 inch and the length 16 times the thickness. Five sample bars shall be fabricated from the test visors for evaluation.

4.7.15.2 Method. Each specimen shall be tested for flexural strength in accordance with ANSI/ASTM D 790, procedure A, method I. The samples shall be mounted on the supports with the convex surface facing up. While the sample is loaded, a load-deflection curve is generated. The test bar shall be stressed until cracking occurs or until the maximum fiber strain of 5 percent is reached. The five samples shall meet the requirements of 3.5.14.

4.7.16 Impact resistance. A 5/8-inch steel ball weighing approximately 0.56 ounce shall be dropped from a height of 50 inches upon the upper horizontal surface of the visor. The steel ball shall strike within the 5/8-inch diameter (geometric center) of the visor. The visor itself shall be supported by a rigid acrylic plastic or steel tube 1 ± 1/8 inch on the inside diameter, 1-1/4 ± 1/8 inches on the outside diameter and approximately 1-inch in height. The top edge of the tube shall have a 1/8 inch neoprene gasket, having a hardness of 40 ± 5 on the durometer scale and a minimum tensile strength of 1200 pounds as determined by ANSI/ASTM D 412, with a minimum elongation of 400 percent as determined by ANSI/ASTM D 412. The diameter of the visor support may be modified so that the 1/8 inch neoprene gasket may support the visor at its periphery. The ball may be guided in its fall by being dropped through a tube, extending to within 4 inches of the visor. The visor shall meet the requirements specified in 3.5.15.

4.7.17 Penetration resistance. The visor shall be supported on a wooden block of such size and shape as to fit the visor securely. A pointed projectile of suitable size, consisting of a new Singer number 25, size 135 x 17 needle, fastened into a holder weighing approximately 1.56 ounces, shall be freely dropped, pointed downward, from a height of 50 inches onto the horizontal outer surface of the visor. The projectile may be guided, but not restricted, in its fall by being dropped through a tube extending to within approximately 4 inches of the visor. The visor shall meet the requirements specified in 3.5.16.

5. PACKAGING

5.1 Preservation. Preservation shall be level A or C, as specified by the acquiring activity (see 6.2.1f).

5.1.1 Level A. Each visor shall be wrapped in flannel and packaged snugly in a unit container conforming to PPP-B-676.

5.1.2 Level C. Each visor shall be preserved and packaged to afford the minimum degree of protection necessary to prevent deterioration or damage during shipment under normal environmental conditions and commercial modes of transportation.

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5.1.3 Precautionary notice. The following notice shall be printed on white paper in accordance with MIL-STD-130 and placed in the package with each lens:

> CAUTION NOTICE: This lens is not for sun protection. It is designed to protect against the possible harmful effects of neodymium laser energy at a wave length of 1064 nanometers (nm). In order to provide the desired protection it must be cared for and used properly. Clean in accordance with instructions in NAVAIR 13-1-6.7 manual.
>
> DO NOT USE FOR PROTECTION AGAINST LASERS OTHER THAN NEODYMIUM (1064 nm)

5.2 Packing. Packing shall be level A, B or C, as specified by the acquiring activity (see 6.2.1f). Shipping containers shall be uniform in size and shape and of minimum cube and tare weight.

5.2.1 Level A. The visors, packaged in the cartons as specified in 5.1.1, shall be packed in fiberboard containers conforming to PPP-B-636, type CF, weather-resistant class, variety SW, grade V3c. Closure and strapping of the containers shall be in accordance with the container specification.

5.2.2 Level B. Visors, packaged as specified in 5.1.1, shall be packed in fiberboard containers conforming to PPP-B-636, type CF, domestic class, variety SW, grade V3c. Closure and strapping of the containers shall be in accordance with the container specification. The gross weight of the packed shipping container shall not exceed the weight limitation of the container specification.

5.2.3 Level C. The visors, which require packing for acceptance by the carrier, shall be packed within exterior type shipping containers in a manner that will insure safe transportation at the lowest rate to the point of delivery. The shipment shall conform to the minimum requirements of the rules and regulations applicable to the mode of transportation selected.

5.3 Marking. In addition to any special markings required by the contract or order (see 6.2.1g), the unit interior and exterior containers shall be marked in accordance with MIL-STD-129 and shall include the date of manufacture (month and year).

6. NOTES

6.1 Intended use. The visors covered by this specification are intended for use with the PRK-37/P helmet shell assembly and the APH-6 helmet shell assembly.

6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number and date of this specification.

b. Government part number and quantity desired.

c. Applicable drawings, including revisions (see 3.4).

d. Whether first article inspection is waived (sec 4.4.1).

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e. Name and address of the first article inspection laboratory (see 4.4.1), and the name of the Government activity responsible for conducting the first article inspection program (see 4.4.1.1).

f. Selection of applicable levels of preservation, packaging and packing (see 5.1 and 5.2).

g. Whether any special markings are required (see 5.3).

h. Items of data required (see 6.2.2).

6.2.2 Data requirements. When this specification is used in an acquisition which incorporates a DD Form 1423, Contract Data Requirements List (CDRL), the data requirements identified below shall be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the approved CDRL incorporated into the contract. When the provisions of DAR 7-104.9 (n) (2) are invoked and the DD Form 1423 is not used, the data specified below shall be delivered by the contractor in accordance with the contract or purchase order requirements. Deliverable data required by this specification is cited in the following paragraph.

Paragraph no.	Data requirement title	Applicable DID no.
4.4.1.1	First article inspection reports	DI-T-5329 Inspection test reports

(Copies of data item descriptions required by the contractors in connection with specific acquisition functions should be obtained from the Naval Publications and Forms Center or as directed by the contracting officer.)

6.3 Qualification. With respect to products requiring qualification, awards will be made only for products which are, at the time set for opening of bids, qualified for inclusion on the applicable Qualified Products List (QPL-22272) whether or not such products have actually been so listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or purchase orders for the products covered by this specification. The activity responsible for the Qualified Products List is the Commander, Naval Air Systems Command, Department of the Navy, Washington, DC 20361; however, authorization for qualification of products shall be obtained from the Commander, Naval Air Development Center, Warminster, PA 18974, (Code 6031). Prior to submission of the samples for qualification inspection, the manufacturer shall submit a request to the Naval Air Development Center (Code 6031) indicating a date on which the samples can be forwarded and also request an authorization number to accompany the samples.

6.4 First article. When a first article inspection is required, the item will be tested and should be a preproduction sample. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, test and approval of the first article.

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6.5 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Preparing activity:
Navy - AS

(Project No. 8475-N176)

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TABLE I. Qualification inspection.

Inspection	Paragraph	
	Requirement	Test method
Visual and dimensional examination		4.7.1
Light transmittance	3.5.1	4.7.2
Neutrality	3.5.2	4.7.3
Chromaticity	3.5.3	4.7.4
Diffuse transmittance	3.5.4	4.7.5
Ultraviolet transmittance	3.5.5	4.7.6
Infrared transmittance	3.5.6	4.7.7
Prismatic deviation	3.5.7	4.7.8
Spherical and cylindrical power	3.5.8	4.7.9
Critical vision area distortion	3.5.9	4.7.10
Wind blast resistance	3.5.10	4.7.11
Haze	3.5.11	4.7.12
Abrasion resistance	3.5.12	4.7.13
Adhesion of coating	3.5.13	4.7.14
Crack propagation resistance	3.5.14	4.7.15
Impact resistance	3.5.15	4.7.16
Penetration resistance	3.5.16	4.7.17

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TABLE II. Sample size, acceptance criteria, tests and examination of the visors.

Inspection	Method paragraph	Sample size	Acceptance criteria
Visual and dimensional examination	4.7.1	Every unit for critical defects. Inspection level II for minor defects.	Reject all units with any critical defect and an acceptable quality level of 6.5 defects per 100 units for minor defects.
Light transmittance	4.7.2	Inspection level II 1/	Acceptance number zero, rejection number 1
Infrared transmittance	4.7.7	Every unit	Reject all defective units
Critical vision area distortion	4.7.10	Inspection level II 1/	Acceptance number zero, rejection number 1
Packaging and identification	4.7.1.1	Every unit for critical defects. Inspection level II for minor defects.	Reject all units with any critical defect and an acceptable quality level of 2.5 defects per 100 units for minor defects.

1/ The sample size shall be based only on the applicable sample size code letter corresponding to the specified inspection level of MIL-STD-105.

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TABLE III. Classification of defects for visual and dimensional examinations.

Defect	Critical	Minor
Material imperfections - foreign matter embedded, striae, waviness, cloudiness, pits, bubbles or scratches:		
Critical vision areas	X	
Noncritical vision areas		X
Surface - unclean, rough, misaligned, or containing cracks, nicks, or other flaws:		
Critical vision area	X	
Noncritical vision areas		X
Dimensions - any dimension not as specified herein or on the applicable drawing	X	
Crack, hole, chip or break	X	
Evidence of warpage	X	
Stain or discoloration not readily removed with water:		
Critical vision area	X	
Noncritical vision area		X
Sharp edge or burr		X
Design or material not as specified	X	

TABLE IV. Classification of defects for packaging and marking.

Defect	Critical	Minor
Package contains more than one visor	X	
Exterior or interior identification missing, incorrect, incomplete, illegible, of improper size, location, sequence or method of application	X	
Permanent marking on visor not as specified on applicable drawing	X	

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TABLE V. Deviations of spectral transmittance. 1/

Wave length nm	Transmittance T 2/	Band n	Wave length range	Average transmittance Tn 3/, 4/	Percent deviation 100(1-Tn/Tc) 5/	Weight	Product
430	0.114						
440	0.118						
450	0.127						
460	0.137	1	430-490	0.133	14	5	70
470	0.142						
480	0.144						
490	0.145	2	460-520	0.145	7	10	70
500	0.147						
510	0.149						
520	0.151	3	490-550	0.151	3	10	30
530	0.153						
540	0.154						
550	0.155	4	520-580	0.155	0	10	0
560	0.157						
570	0.158						
580	0.159	5	550-610	0.159	2	10	20
590	0.160						
600	0.160						
610	0.160	6	580-640	0.160	2	10	30
620	0.161						
630	0.161						
640	0.160	7	610-670	0.160	3	10	30
650	0.159						
660	0.159						
670	0.158	8	640-700	0.158	2	5	10
680	0.157						
690	0.156						
700	0.153	9	670-730	0.153	1	1	1
710	0.151						
720	0.149						
730	0.148						
					Totals 6/	71	261

1/ The glass used for this worked example is Bausch and Lomb neutral N-15.

2/ T = Transmittance at 10 nanometer intervals.

3/ Tn = Average transmittance of 60 nanometer band.

4/ The average transmittance, Tn, for a given band is the average of the seven tabulated values within that band, except that the first and the last values are divided by 2, and the average computed by dividing the sum of the values by 6.

5/ Daylight transmittance, Tc = 0.155.

6/ Average percentage deviation of spectral transmittance within nine spectral bands from daylight transmittance, i.e. Average percent deviation = 261/71 = 3.7 percent.

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TABLE VI. Computation table of coordinates. 1/, 6/

Wave length (nanometers)	$\bar{x}$	$\bar{y}$	$\bar{z}$	2/ T	3/ X	3/, 4/ Y	3/ Z
380	4	0	20	0.104	0	0	2
390	19	0	89	0.240	5	0	21
400	85	2	404	0.301	26	1	122
410	329	9	1,570	0.275	90	2	432
420	1,238	37	5,949	0.174	215	6	1,035
430	2,997	122	14,628	0.110	330	13	1,609
440	3,975	262	19,938	0.093	370	24	1,854
450	3,915	443	20,638	0.092	360	41	1,899
460	3,362	694	19,299	0.100	336	69	1,930
470	2,272	1,058	14,972	0.110	250	116	1,647
480	1,112	1,618	9,461	0.122	136	197	1,154
490	363	2,358	5,274	0.132	48	311	696
500	52	3,401	2,864	0.140	7	476	401
510	89	4,833	1,520	0.142	13	686	216
520	576	6,462	712	0.142	82	918	101
530	1,523	7,934	388	0.141	215	1,119	55
540	2,785	9,149	195	0.141	393	1,290	27
550	4,282	9,832	86	0.155	664	1,524	13
560	5,880	9,841	39	0.170	1,000	1,673	7
570	7,322	9,147	20	0.167	1,223	1,528	3
580	8,417	7,992	16	0.153	1,288	1,223	2
590	8,984	6,627	10	0.142	1,276	941	1
600	8,949	5,316	7	0.136	1,217	723	1
610	8,325	4,176	2	0.136	1,132	568	0
620	7,070	3,153	2	0.137	969	432	0
630	5,309	2,190		0.137	727	300	
640	3,693	1,443		0.138	510	199	
650	2,349	886		0.150	352	133	
660	1,361	504		0.199	256	94	
670	708	259		0.270	191	70	
680	369	134		0.368	136	49	
690	171	62		0.475	81	29	
700	82	29		0.576	47	17	
710	39	14		0.620	24	9	
720	19	6		0.636	12	4	
730	8	3		0.643	5	2	
740	4	2		0.642	3	1	
750	2	1		0.632	1	1	
760	1	1		0.620	1	1	
770	1	0		0.600	1	0	
				Totals 5/	13,992 (X)	14,790 (Y)	13,228 (Z)

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TABLE VI. Computation table of coordinates. 1/, 6/ - Continued

1/ The glass used for this worked example is Bausch and Lomb neutral glass.

2/ The symbol "T" represents spectral transmittance, the ratio of transmitted to homogeneous radiant flux.

3/ $X = \bar{x}T$, $Y = \bar{y}T$, $Z = \bar{z}T$

4/ Daylight transmittance, Tc, where: Tc = Y/1,000 = 14.8 percent

5/ Chromaticity coordinates:
$x = X/(X + Y + Z) = 13,992/(13,992 + 14,790 + 13,228) = 0.3331$
$y = Y/(X + Y + Z) = 14,790/(13,992 + 14,790 + 13,228) = 0.3521$

6/ Daylight transmittance, Tc, and chromaticity coordinates, x and y, for standard illuminant C.

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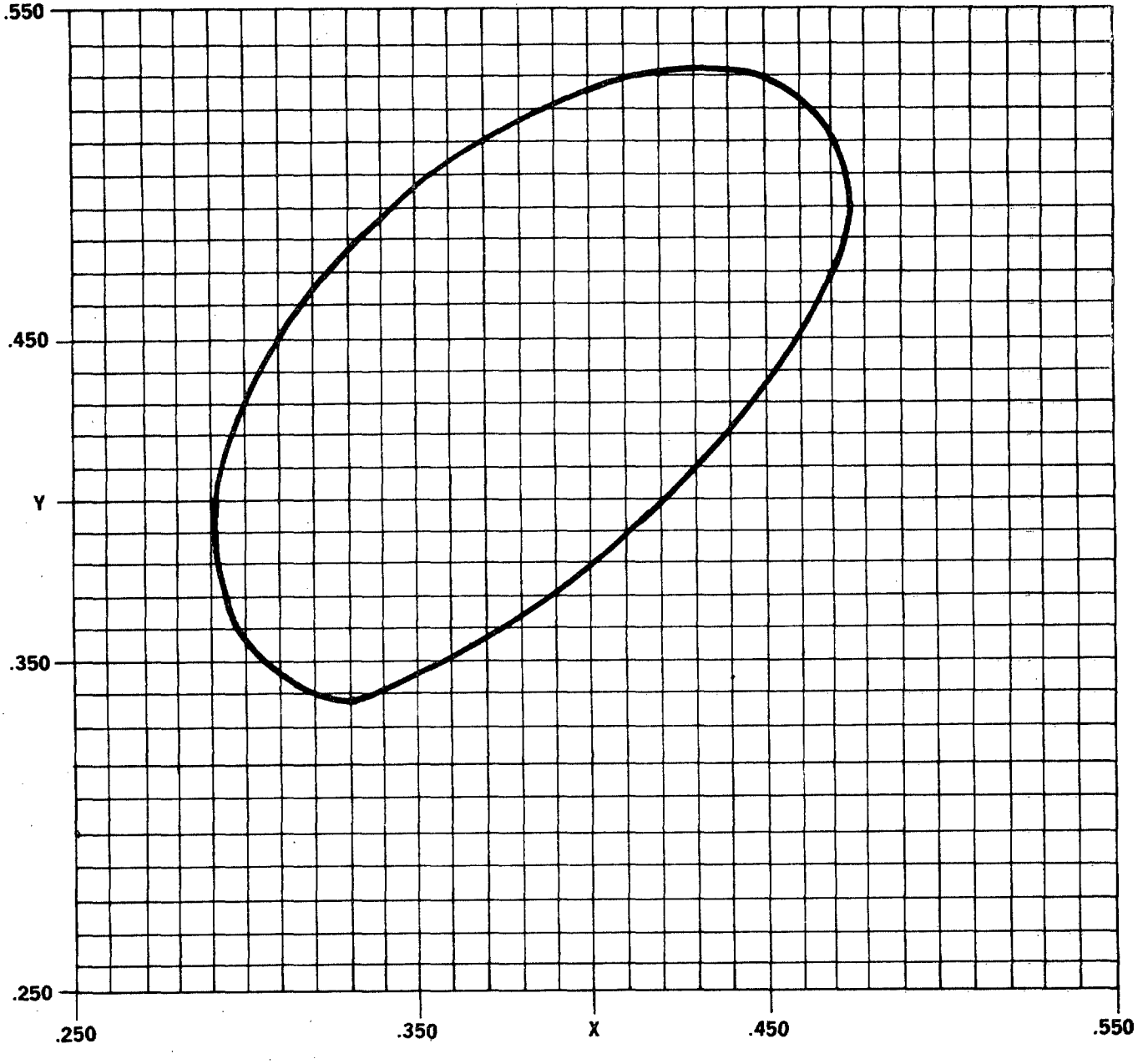
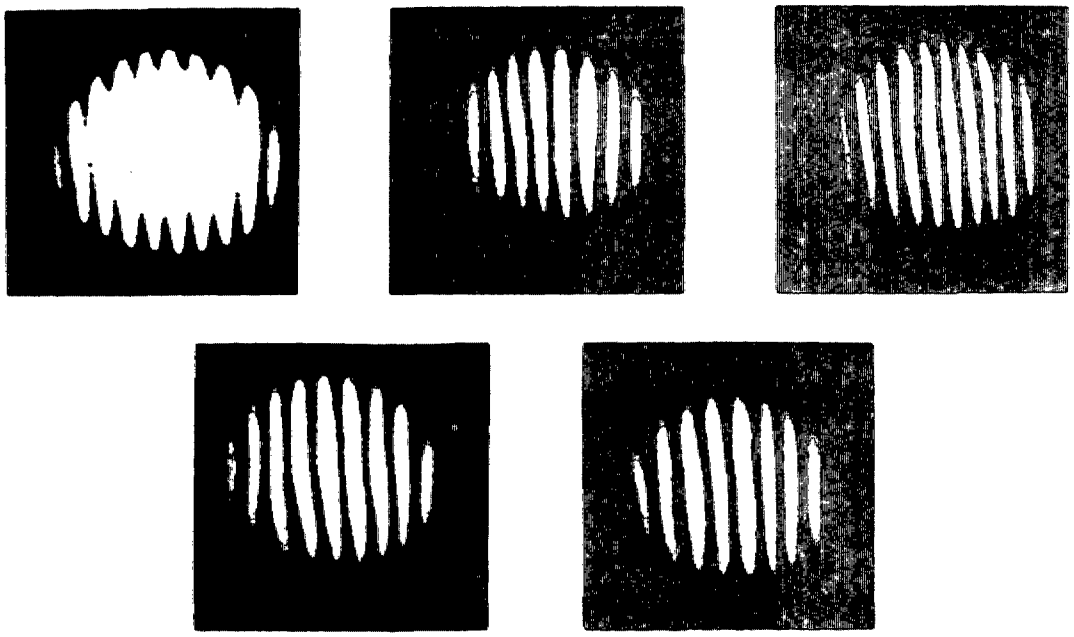


FIGURE 1. Chromaticity coordinates.

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PILOT VISOR
DISTORTION STANDARDS

ACCEPTABLE



UNACCEPTABLE

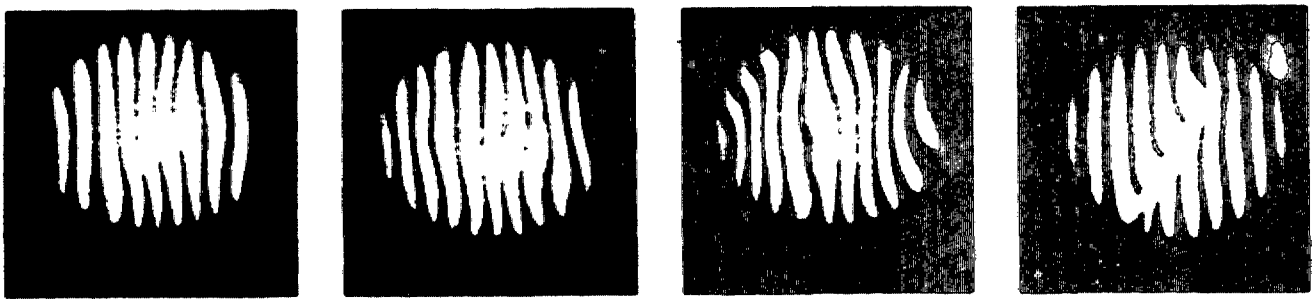
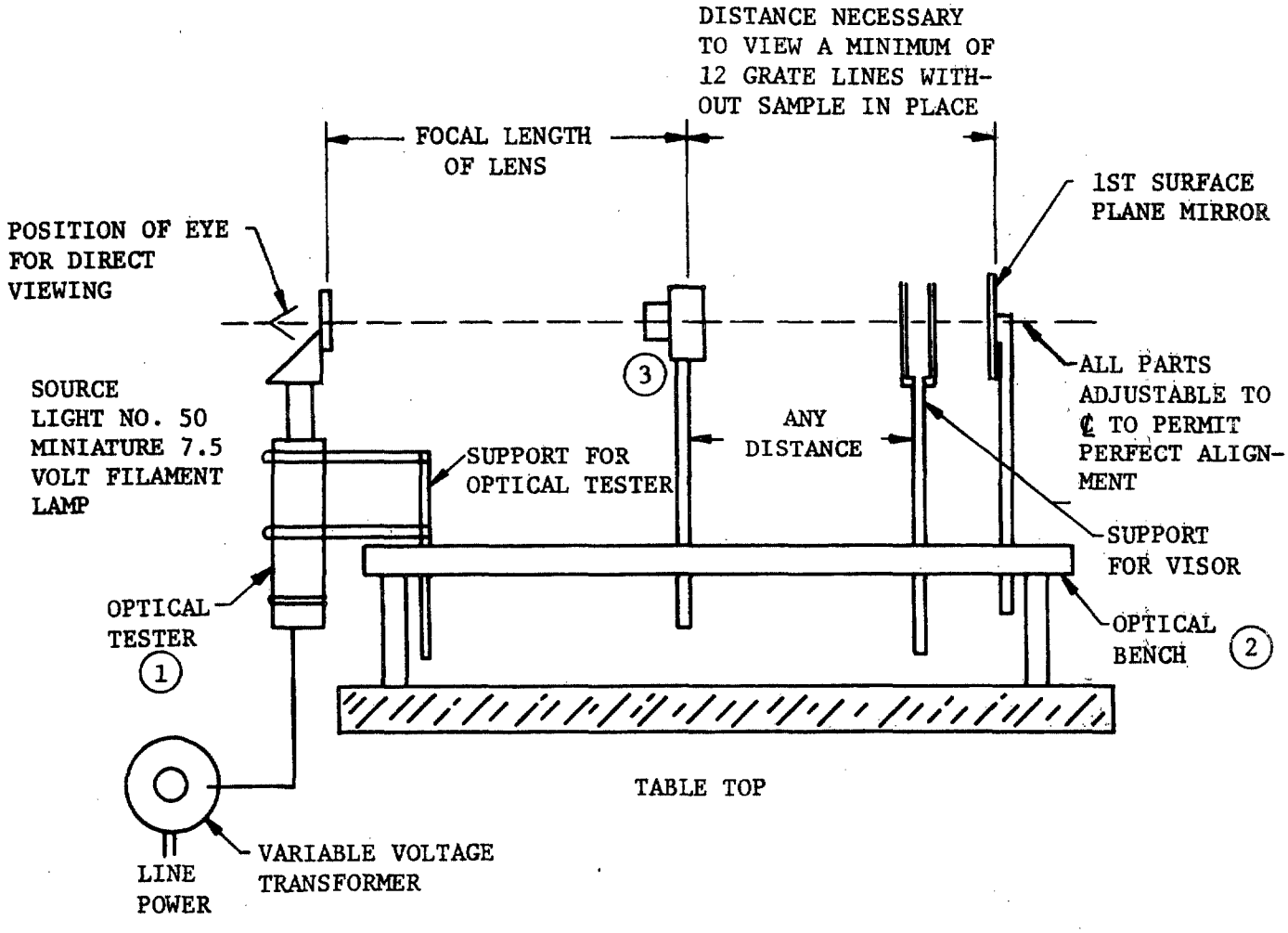


FIGURE 2. Distortion standards.

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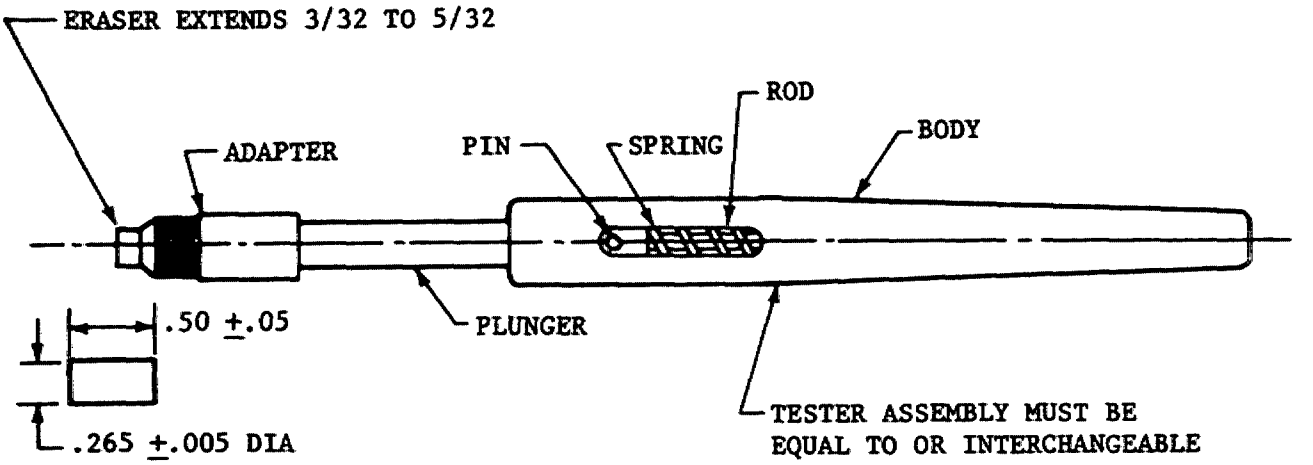
1. MODEL "B" OPTICAL TESTER WITH A 60-LINE GRATING, WITH MODEL "O" OPTICAL BENCH ADAPTER OR EQUIVALENT. (OBTAINABLE FROM ANN ARBOR OPTICAL CO., P.O. BOX 2056, ANN ARBOR, MICHIGAN).

2. OPTICAL BENCH OBTAINABLE FROM CENTRAL SCIENTIFIC CO., OR ITS EQUIVALENT OBTAINABLE FROM ANY EQUIPMENT SUPPLIER.

3. TELEPHOTO CAMERA LENS - A SCHNEIDER TELE-ARTON 1:5.5/240 mm, NORMAL COLOR COATED, BARREL ONLY OR EQUIVALENT.

FIGURE 3. New London-Ann Arbor distortion tester.

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ERASER PLUG PER
MIL-E-12397

SUGGESTED SOURCE:
SUMMERS LAB INC.
BOX 162
FORT WASHINGTON, PA 19034

DIMENSION IN INCHES.

FIGURE 4. Eraser abrasion tester assembly and eraser plug.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

DD FORM 82 MAR 1426 PREVIOUS EDITION IS OBSOLETE.

(TO DETACH THIS FORM, CUT ALONG THIS LINE.)

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